Data Collection – Process and Steps

Introduction

Data collection is the process of gathering information in a systematic way to answer research questions, solve problems, or make decisions.
The collected data may be **numerical (quantitative)** or **descriptive (qualitative)**.

Good data collection is important because:

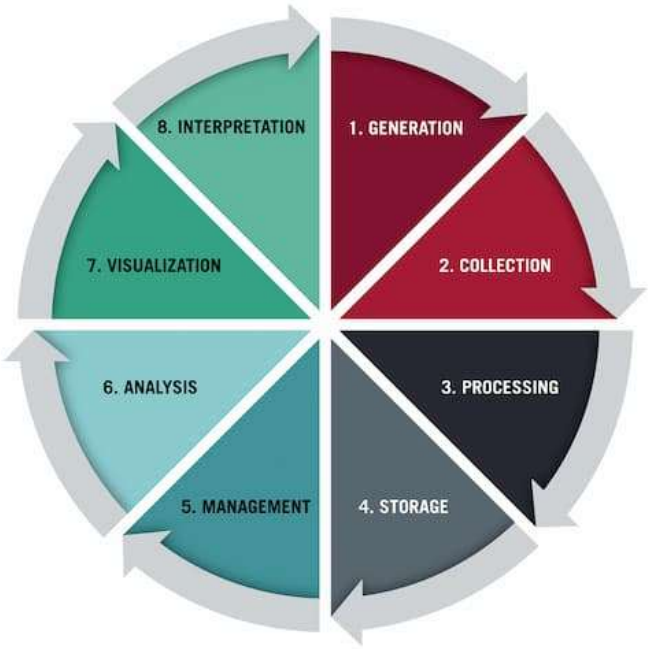
- It improves accuracy.
- It supports correct conclusions.
- It reduces errors and bias.
- It helps in proper analysis.

Examples:

- Survey responses from students
- Sales records of a company
- Temperature readings
- Interview answers

Steps in the Data Collection Process

The data collection process follows a series of well-defined steps.



1. Define the Objective

First, clearly decide **why** data is needed.

Questions to ask:

- What problem am I studying?
- What information is required?
- Who will use this data?

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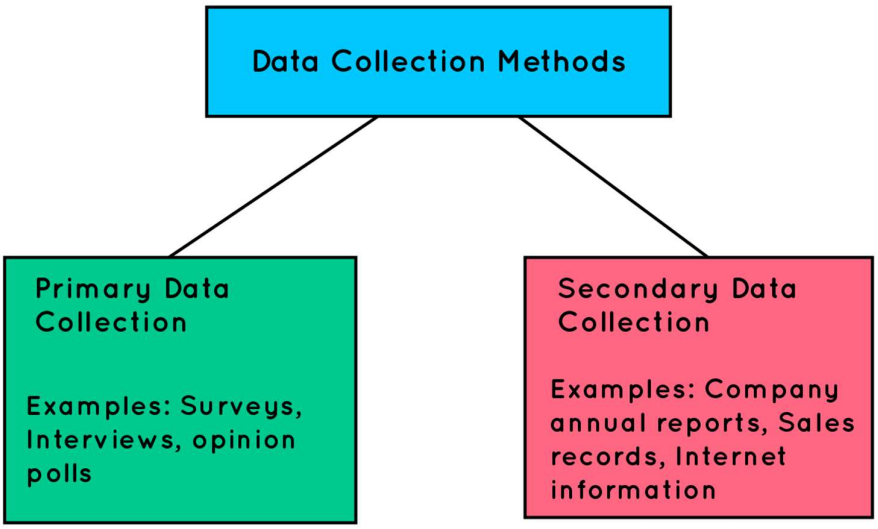
2. Identify the Type of Data Needed

Decide what kind of data is required:



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3. Select the Data Collection Method

Choose suitable methods based on the objective.

Common methods:

- **Questionnaires / Surveys** – Written questions answered by people.
- **Interviews** – Asking questions face-to-face or online.
- **Observation** – Watching behavior or events.
- **Experiments** – Testing in controlled conditions.
- **Documents and Records** – Using existing files and reports.

4. Design the Data Collection Tools

Prepare instruments used to gather data.

Examples:

- Questionnaire form
- Interview schedule
- Observation checklist

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5. Collect the Data

Now gather the actual information.

During this step:

- Follow the plan carefully.
- Record answers correctly.
- Be polite to respondents.
- Maintain confidentiality.
- Avoid influencing answers.

Accuracy is very important at this stage.

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6. Organize and Record the Data

After collection, arrange the data properly.

- Sorting responses
- Coding answers
- Entering into tables or spreadsheets
- Removing incomplete data

This makes analysis easier.

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7. Check Data Quality

Verify whether the data is:

- Complete

- Correct
- Consistent
- Reliable

Errors or missing values should be corrected if possible.

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8. Store and Secure the Data

Keep the collected data safe.

Methods:

- Password-protected files
- Locked cabinets for paper forms
- Backups in storage devices

Security is important to protect privacy.

Decision Making and Steps in Business Analytics

Decision making in business analytics is the process of selecting the best possible action by analysing data and interpreting results in a business context.

Instead of guessing, managers use:

- Reports and dashboards
- Data summaries
- Forecasts
- Statistical models
- Optimization techniques

The aim is to move from **data → information → insight → action**.

Types of Decisions in Business

Business analytics supports different levels of decisions:

1) Strategic Decisions

- Long-term and high impact
- Taken by top management
- Concern company direction
- Example: entering a new market or acquiring another firm

2) Tactical Decisions

- Medium-term
- Related to departmental planning
- Example: deciding marketing budget for next quarter

3) Operational Decisions

- Short-term and routine
- Day-to-day activities
- Example: deciding reorder quantity for inventory

Steps in Business Analytics for Decision Making

Business analytics follows a structured process to convert raw data into decisions.

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1. Define the Business Problem

This is the most important step. The problem must be stated clearly.

It should answer:

- What exactly is wrong or needs improvement?
- What decision must be made?
- What is the goal?
- Are there constraints such as budget, time, or rules?

Example:
Sales are falling in one region → identify causes and suggest actions to increase sales.

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2. Understand the Business Context

Analysts study the background of the problem:

- Company objectives
- Industry conditions
- Competitors
- Regulations
- Customer expectations

This ensures the analysis focuses on what really matters to the business.

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3. Identify Data Requirements

Decide what information is required to solve the problem.

This includes:

- What variables are needed? (sales, cost, demand, customer age, etc.)
- Time period (last year, last five years)

- Level of detail (daily or monthly data)
- Data format

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4. Collect the Data

Data may come from many sources:

- Internal systems (sales databases, finance records, CRM, ERP)
- Surveys and feedback forms
- Website logs and mobile apps
- Sensors and machines
- Government or public data

The data may be:

- **Structured** – tables, spreadsheets
- **Unstructured** – text reviews, images, emails

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5. Clean and Prepare the Data

Raw data is usually messy and cannot be analyzed directly.

Preparation includes:

- Removing duplicates
- Handling missing values
- Correcting wrong entries
- Standardizing units and formats
- Combining data from different sources

This step often takes much time but is essential for reliable results.

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6. Explore the Data

Here analysts try to understand the data before building models.

Techniques include:

- Calculating averages and totals

- Creating charts and graphs
- Studying trends and seasonality
- Finding relationships between variables

This step may reveal unexpected patterns or problems in the data.

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7. Apply Analytical Models

Based on the objective, suitable analytical methods are used:

Descriptive Analytics – What happened in the past?

Sales reports, dashboards.

Diagnostic Analytics – Why did it happen?

Root-cause analysis, correlations.

Predictive Analytics – What is likely to happen in the future?

Forecasting, regression models.

Prescriptive Analytics – What should be done?

Optimization models, simulation.

These models transform data into useful insights.

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8. Make the Decision

Decision makers compare possible options using the insights obtained.

They consider:

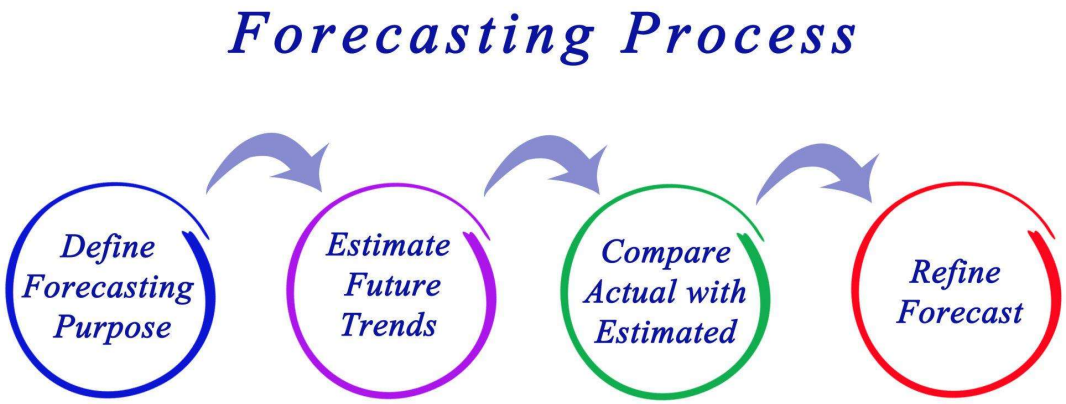
- Business objectives
- Financial impact
- Risk level
- Feasibility

Forecasting

Forecasting is the process of predicting future events by analyzing past and present data.
In business, forecasting is widely used to estimate:

- Future sales
- Customer demand
- Revenue and profit
- Inventory levels
- Market growth

Good forecasting helps managers plan better, reduce risk, and make informed decisions.



Importance of Forecasting

Forecasting is important because it helps in:

- Production planning
- Budget preparation
- Staffing decisions
- Marketing strategies

- Capacity expansion
- Risk management

Without forecasting, businesses may overproduce, underproduce, or invest wrongly.

Steps in the Forecasting Process

Forecasting follows a systematic sequence of steps to obtain reliable predictions.

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1. Define the Objective

First, clearly state **what is to be forecast** and **why**.

Questions to consider:

- Are we forecasting sales, demand, or revenue?
- For which product or region?
- For short-term or long-term period?

Example:
Forecast monthly sales for the next one year.

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2. Determine the Time Horizon

Decide the length of the forecast:

- **Short-term** – days or months (inventory control)
- **Medium-term** – months to 2 years (production planning)
- **Long-term** – several years (expansion or investment)

The time horizon affects the method chosen.

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3. Identify Data Requirements

Decide what data is needed:

- Past sales figures
- Customer records
- Economic indicators

- Seasonal information
- Promotions or price changes

Accurate data leads to better forecasts.

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4. Collect the Data

Gather data from different sources:

- Company records
- Accounting systems
- Market research
- Government statistics
- Industry reports

The data should be reliable and relevant.

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5. Clean and Prepare the Data

Before analysis, the data must be prepared.

This includes:

- Removing duplicate records
- Handling missing values
- Correcting errors
- Converting units
- Organizing data into tables

Clean data improves forecasting accuracy.

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6. Select a Forecasting Method

Choose a suitable technique based on:

- Nature of data
- Time horizon
- Accuracy required

- Cost and complexity

Common methods:

Qualitative Methods

Used when little data is available.

- Expert opinion
- Delphi method
- Market surveys

Quantitative Methods

Use mathematical models.

- Moving averages
- Exponential smoothing
- Time-series models
- Regression analysis

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8. Build the Forecast Model

Apply the selected method to the data.

Activities include:

- Fitting the model
- Calculating parameters
- Generating future values

This step produces the forecast results.

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9. Validate the Forecast

Check how accurate the model is by:

- Comparing predicted values with actual past values
- Calculating error measures
- Adjusting the model if required

This improves reliability.

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10. Present and Use the Forecast

Show results clearly using:

- Tables
- Charts
- Reports
- Dashboards

Managers then use the forecast for planning and decisions.